

INNOVATION DAY 2026

SALLE DES COMITÉS
VÉLIZY CREATIVE CENTER

TUESDAY, OCTOBER 13
09:30 AM – 03:00 PM

Connectivity

ECU

Displays-HUD

**ADAS/AD Sensors
and Features**

Interaction Modalities

Chassis

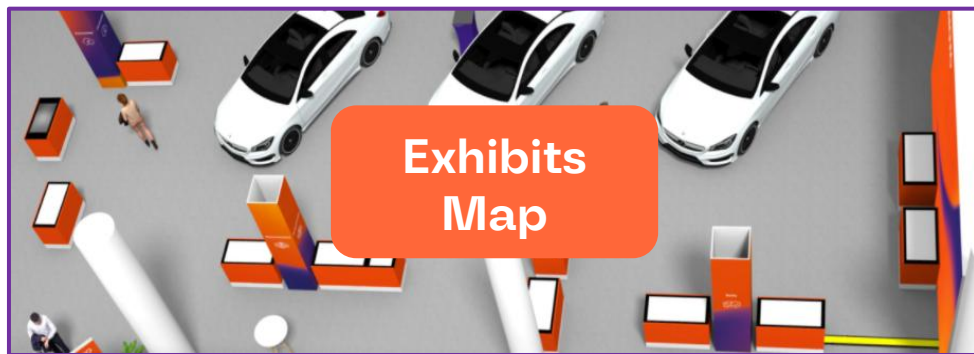
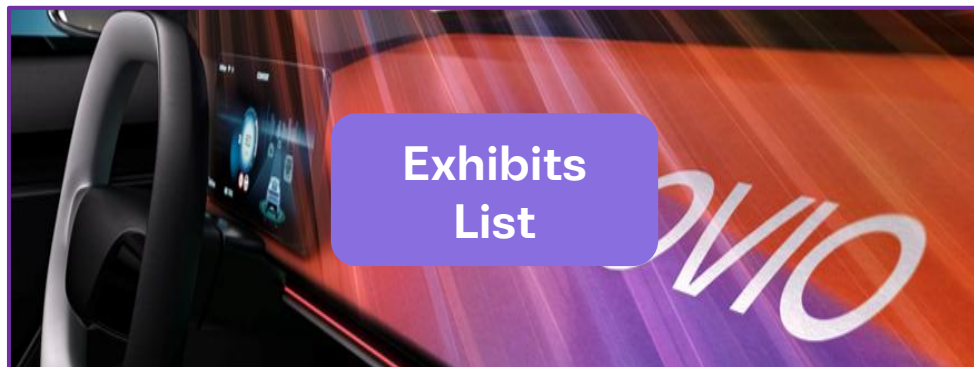
Lighting

Sound System

The Adaptive Powerhouse for Future Mobility

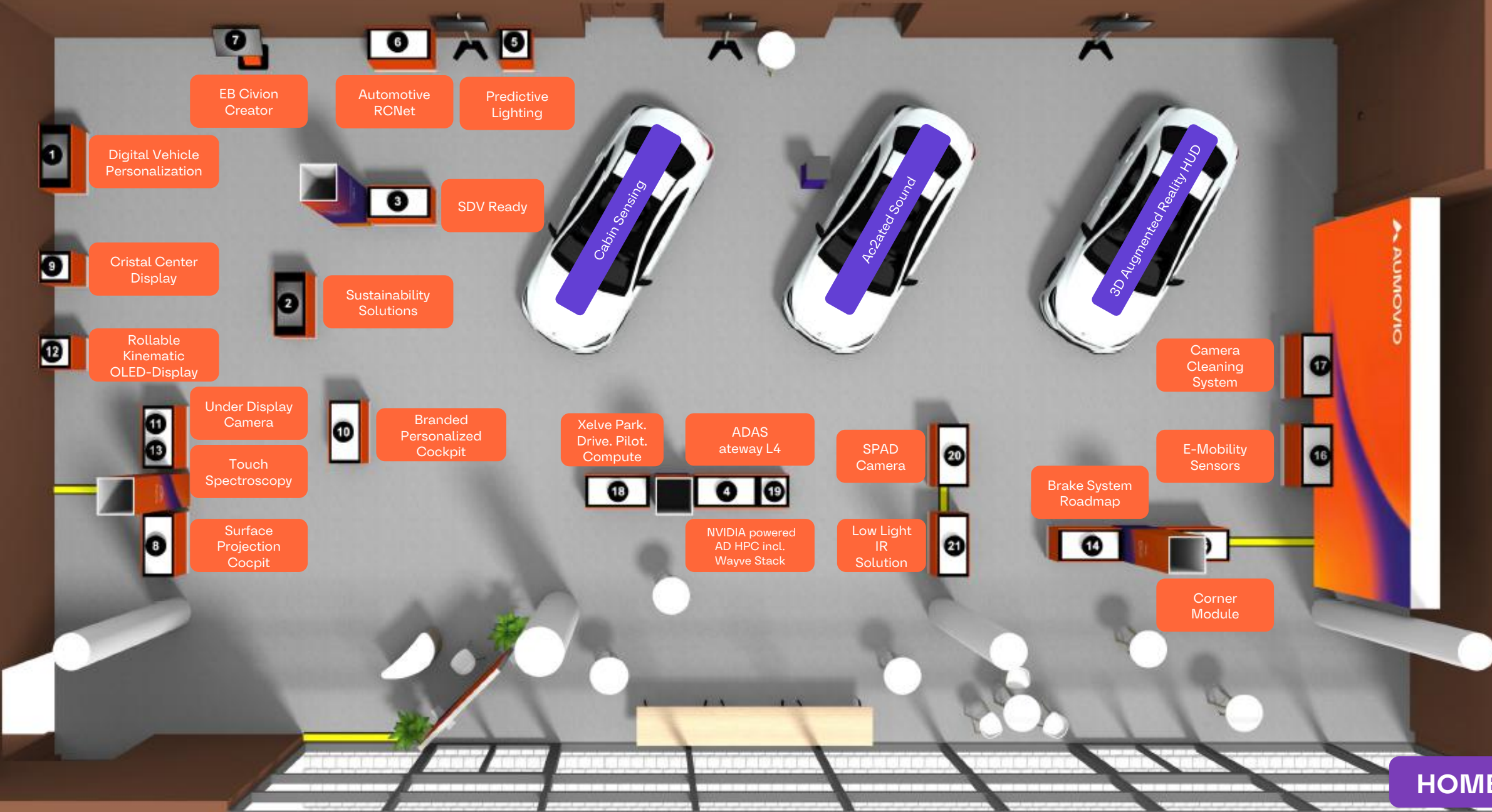
Aumovio Innovation Day at Stellantis Creative Center Velizy

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ADAS ateway L4

NVIDIA powered AD HPC incl. Wayve Stack

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4

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SPAD Camera

Low Light IR Solution

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Brake System Roadmap

Camera Cleaning System

E-Mobility Sensors

Corner Module

17

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Cabin Sensing

Ac2ated Sound

3D Augmented Reality HUD

AUMOVIO



**SOP
2030-2032**

Benefits & Features:

- AI-powered personalization
- Full-environment customization: ambient lighting, sounds, animated display backgrounds
- Multi-input control: voice, text, audio cues, and camera capture
- Future-ready: extendable to XR glasses

Technology:

- Generative AI cloud service
- Real-time orchestration of HMI elements
- Cloud-native
- Mixed Reality (XR) integration

Sustainability Solutions

Create value for end-of-life mechatronic products



**SOP
2026+**

Benefits & Features:

- **Remake service** (Reman, Repair)
 - Reduce or avoid LBO for aftersales
 - Lower spare part price for end-customer
 - Carbon footprint reduced vs new part
- **Revive service** (Reuse, Recycle)
 - Reduce need of critical material extraction
 - Sovereignty: enable new (local) supply source for electronic material.

... supports ELVR regulation

Technology:

- Industrial Process for Remanufacturing
- Reverse Engineering analysis, ECU factory reset
- Data management for product traceability & decision making (DPP, AI)
- Product and PCB automatic disassembly
- Smart sorting (Visual inspection,...)

SDV Ready Open Telematics Platform



SOP
From Now up to
2027 & 2028

Benefits & Features:

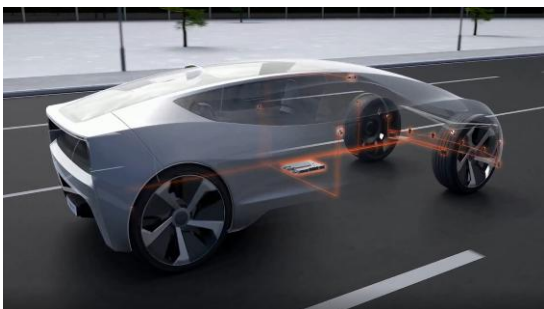
- Reuse of software across hardware platforms reduces redevelopment effort
- HW-independent architecture enables quicker rollout of new features and updates
- One SW foundation spanning 5G, 5G Advanced, and future 6G platforms.
- Multi-modem architecture delivers high bandwidth, redundancy, and resilience for advanced AD applications.
- OEM-controlled, cloud-agnostic multi-ECU update solution with competitive total cost of ownership.

Technology:

- SDV-ready telematics platform separates application and hardware layer
- Roadmap from low-cost 5G solutions to 5G advanced and future 6G-ready architectures
- Multi-modem redundancy platform
- Satellite connectivity support
- Cloud-agnostic multi-ECU OTA solution

Fast lane to L4 – NVIDIA powered AD HPC incl. Wayve Stack

4



**SOP
2027**

Benefits & Features:

- Standardized Nvidia-based AD HPC platform supporting automated driving functions from L2 – 4
- Pre-integrated sensors, software, and system services that accelerate time to market
- Flexible integration of industry-leading AI driving technologies such as Wayve.
- Turnkey, end-to-end capability— from hardware and software to integration, validation, calibration, homologation, and vehicle deployment.

Technology:

- Fully operational hardware development kits with pre-integrated compute, sensors, software, and services enabling the fast lane to automotive series production
- AD HPC powered by NVIDIA Thor for next-generation AI workloads.
- Real-time object detection and classification allowing to receive and process video stream from the second Thor via PCIe
- Hyperion-compatible solutions
- Robust environmental sensing



SOP
Late 2028

Benefits & Features:

- Empowerment of the vehicle central communication gateway for functional safety required for L4 automated driving, using smart redundancy
- Without this, the vehicle would stop in the driving lane in case of a failure. With this gateway, vehicle can continue operation still for 5 min
- AUMOVIO as knowledgeable partner with deep experience in vehicle gateways as leading supplier with long product track record
- In-house experience with automated driving solutions (HPC, stack)

Technology:

- Advanced routing deployment on modern architectures
- Introduction of modern IEEE1722 based solution to transport legacy buses
- Complete automatic toolchain for Generating and assessing of routing
- Log files capturing long time device under test behavior imported and analyzed by Offline analysis tool, statistics generated and abnormal behavior detected

Predictive Lighting Innovative Technologies



SOP:
Function readiness
2027*

depending on available
cloud/ infrastructure

Benefits & Features:

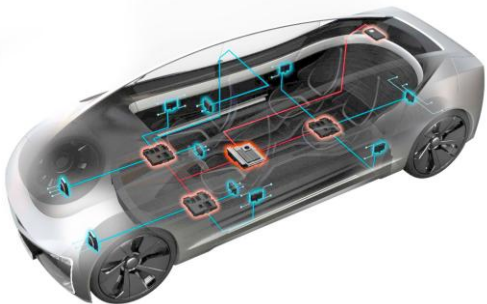
- Safer night driving: more light, less glare
- Sensor fusion boosts lighting safety, availability and performance
- External and onboard sensor fusion gives drivers more reaction time
- Better visibility, path guidance and attention alerts
- HD lighting: from convenience to safety
- Crowd and cloud data enable fast responses to traffic changes
- Cloud sensor fusion saves power in well-lit areas

Technology:

- By combining electronic horizon, HD maps, cloud intelligence and onboard sensor fusion, predictive lighting software improve night lighting across varied conditions and enable rapid responses to sudden traffic events
- HD projection systems enable new lighting functions, such as trajectory guidance, while enhancing existing functions
- Highly scalable ZCUs support a wide range of vehicle architectures

Automotive Remote Control Network (Automotive RCNet)

Innovative networking technology



SOP
Early 2028

Benefits & Features:

- Improved and simplified vehicle architecture
- Enabling easy feature upgrades for the consumer during the life of the car without big impact in the in-car communication/hardware
- Lower peripheral costs and increased modularity, better reuse and scalability by introduction of standardized I/O boxes
- Simplified variant handling and reduced complexity
- Standardization of smart actuator control ECUs and I/O boxes leading to less HW variants and economies of scale

Technology:

- AUMOVIO is developing Automotive RCNet as an open network technology for SDV
- Automotive RCNet brings a standardized communication protocol between application software host and Automotive RCNet devices for true hardware/software decoupling
- Automotive RCNet remote control services using lower-cost CAN/LIN compared to Ethernet



SOP
2026

Benefits & Features:

- **Accelerate time to market** by decoupling cockpit HMI development from physical HW
- **Reduce program cost** with higher productivity, and fewer validation
- **Streamline your branding** and multi variant handling with high reuse capabilities
- Deliver **differentiated, branded cockpit experiences** at the speed the market needs

Technology:

- **Cloud native**, browser based (AWS / Google / On prem)
- Connect your UX design via **Figma** and customize IVI with a **Theming Engine**
- **Emulate your cockpit SW** via **virtual Android targets**
- **Abstract vehicle** data, simulate vehicle signals, test and **validate with no HW**

Surface Projection Cockpit

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**SOP
2027**

Benefits & Features:

- Light Projection enables animated brand, décor and functional experiences e.g.
 - Emotionalizing of vehicle brand
 - Enabling of new brand experiences with magic appearance
 - Canvas for art projection
- Small form factor of projector allows roof integration
- Use of almost all surfaces (dashboard, head liner, ...) possible

Technology:

Light Projection:

- Seamless pillar-2-pillar experience with 2 synchronized projectors
- Hi-Resolution full color RGB graphics

Vehicle Integration & Design

- Color and trim selection can be aligned with projection system to achieve best in-vehicle integration.

Cristal Center Display

In cooperation with Swarovski Mobility



**SOP
2028**

Benefits & Features:

- World's first jewel-shaped touch displays for luxury appearance
- Transparent MicroLEDs in combination with tinted crystal creates a one-of-a-kind user
- Special floating effect of the screen together with the reflections on the complex three-dimensional surface create a novel visual impression

Technology:

- μ LED Display 10"
- New: Touch integration through 10mm thick crystal
- New: Multi finger interaction
- Frameless sculptural body of tinted premium crystal
- MicroLED for high brightness and ultra-high contrast
- Tinted premium crystal with precision facet cut
- New: Ultra thin touch layer for interaction

Branded Personalized Cockpit

10



**SOP
2028**

Benefits & Features:

- **Individualization and new end customer experiences** with Color E Ink decor display increases brand-bonding
- **Invisible camera integration** for driver monitoring with new Automotive Qualified Curved OLED Panel for clean interior design and better user acceptance
- **Distraction-free Entertainment** with Partial Switchable Privacy allows codrivers to watch movies w/o disturbing the driver
- **Integration of 3rd party contents** and constant experience updates via UX assets and our partner banbutsu

Technology:

Color E Ink decor display

- 23,1” reflective color E-ink display (150 ppi)

Under Display Camera

- Infrared DMS Camera seamlessly integrated (invisible)
- Display resolution: 167ppi in entire display area

Partial Switchable Privacy

- Seamlessly integrated switchable privacy feature area
- 34’ OLED Display (179 ppi, 1000 cd/m²)

Under Display Camera – LCD with Edge lit Backlight



**SOP
2029**

Benefits & Features:

- NIR Camera placed under display Active Area
- Edge lit backlight integration design with standard LCD panel
- Integration of nanostructure light guide for homogenous illumination
- Camera compensation SW implemented for better camera performance

Technology:

NIR Camera

- 1,5MP IR Camera behind display
- IR illumination on the left/right side and frame
- Camera image compensation SW

Display

- TFT LCD Panel
- 10.25” inch screen size
- Resolution 1920 x 720 (200ppi)
- Edge Backlight

Rollable Kinematic OLED Display

12



**SOP
2029**

Benefits & Features:

- Product Concept will enable the overall trend of “Display on Demand”
- Variable and freely configurable display movement based on Aumovio’s driving electronics
- Fully automotive qualified rollable OLED display panel available
- Kinematic concept enables high stiffness for high-quality impression while interacting with the display surface

Technology:

- Based on flexible OLED display panel
- Brushless motor technology for most silent and smooth kinematic drive
- Fully automotive qualified OLED display technology available
- Based on strategic partnership with automotive kinematic system supplier
- Rollable display panel allows dynamic display size even under tight cockpit packaging requirements

Touch Spectroscopy (Alcohol Sensing)

UDC incl. dot projector & touch spectroscopy



**SOP
2030**

Benefits & Features:

- NIR Camera placed under display Active Area
- Active driver-monitoring functions for real-time assessment

NIR Dot-Projector

- Seat belt detection
- 3D depth map
- Vital signs

Touch Spectroscopy Sensor

- Alcohol & Moisture sensing

Technology:

NIR Camera

- 1.5MP IR Camera behind display
- IR illumination on left/right side besides display
- Special algorithm for image enhancement enables camera placement behind active area

Display

- Innovative automotive qualified OLED panel with special pixel design enhances NIR transmission rate
- Flat 12.6" OLED display
- Resolution 2880 x 1800 (270ppi)



SOP
2026-2030+

Benefits & Features:

- Cost benefit due to enhanced driving dynamics with longer EV range and reduced maintenance
- Simplification with modular and scalable across ICE, HEV and BEV platforms
- Innovation: Enables four-wheel parking, eliminates the need for hazardous brake fluid, and supports full autonomous driving functionality.

Technology:

- One-Box: MK Cx
- Distributed: MK Dx
- Semi-Dry Brake System
MK C2 (Front Axle) + EMB (Rear Axle)
- Full-Dry Brake System
4x EMB (Front + Rear)



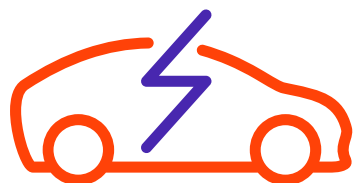
**SOP
2030+**

Benefits & Features:

- Vehicle motion to the next level
- Full steering capability
- Individual steering up to 4 wheels
- New steering functions
e.g. turn-on-spot, crab walking, easy parking, ...
- First Corner Module that fits into a state-of-the-art body

Technology:

- Decoupled Steering
- Software-defined control
- Drive Brake Unit
- Air Spring System
- Plug-and-Play Modular Chassis



SOP
2027-2029

Benefits & Features:

Advanced E-mobility sensor solutions for:

- Better E-motor performance, less weight
- Higher safety
- Improved comfort, NVH behavior & energy consumption

Technology:

- E-Motor Sensors
- Advanced Battery Monitoring for LV and HV Batteries
- Chassis Sensors

Camera Cleaning System

17



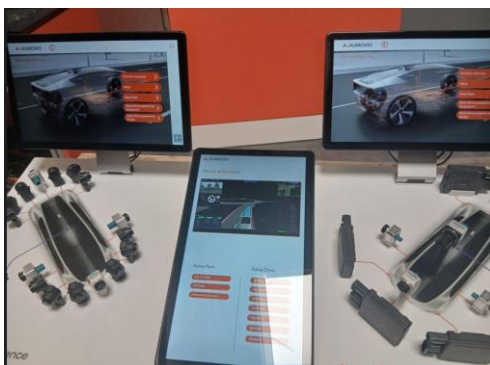
**SOP
2027**

Benefits & Features:

- Increased driver satisfaction
- Prepared for autonomous driving
- Higher sustainability rating
- Increased road safety
- Cleaning with fluid and air (for removing droplets)

Technology:

- Different spray pattern for camera and Lidar
- Static or telescopic nozzles possible
- Heated with heater wire for nozzle and/or lens
- Integrated check valve for short response time



**SOP
2025
With QC 2028**

Benefits & Features:

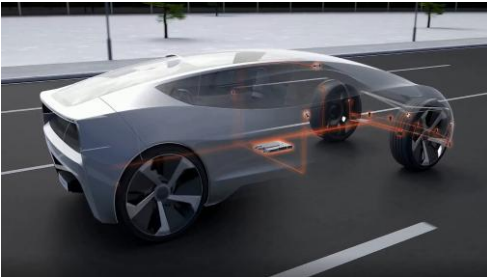
- AUMOVIO delivers Xelve, a scalable solution for both volume and premium markets.
- It seamlessly integrates hardware and automotive software to create an entry point for Software-Defined Vehicles, supporting SAE levels L2 to L4.
- Xelve Drive provides SAE L2(+) system solutions together with our trusted partners—from architecture definition and integration services to data services and vehicle level calibration, triage, validation and homologation services.

Technology:

- **Xelve Park** showcases the years of innovation from the Transparent vehicle features to Radar Vision Parking , while incorporating the redefined user experience enabled by the Human Vision knowledge
- **Xelve Drive** delivers a hybrid architecture to bring the benefits of end-to-end with classical drive policy, while still being cost efficient
- **Xelve Pilot** conveys a new business model (Hardware as a service) and a pioneering technology collaboration

Fast lane to L4 – NVIDIA powered AD HPC

Xelve Drive Powered by Wayve



SOP
Mid 2028

Benefits & Features:

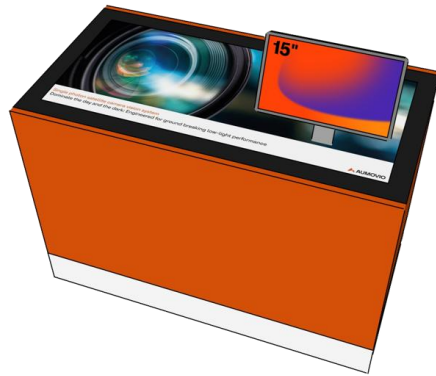
- Standardized Nvidia-based AD HPC platform supporting automated driving functions from L2 – 4
- Pre-integrated sensors, software, and system services that accelerate time to market
- Flexible integration of industry-leading AI driving technologies such as Wayve.
- Turnkey, end-to-end capability— from hardware and software to integration, validation, calibration, homologation, and vehicle deployment.

Technology:

- Fully operational hardware development kits with pre-integrated compute, sensors, software, and services enabling the fast lane to automotive series production
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- Hyperion-compatible solutions
- Robust environmental sensing

Low Light Solution SPAD for Satellite Camera

Single photon satellite camera vision system



SOP
2031/32
Maybe Earlier

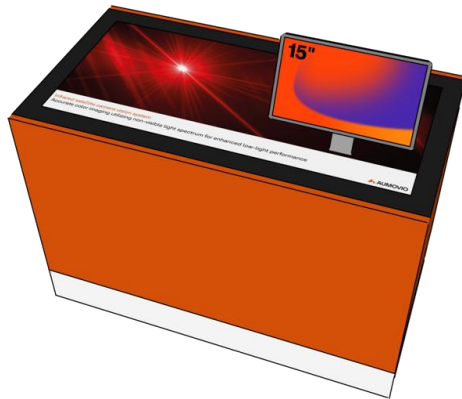
Benefits & Features:

- Day & night imaging from bright sunlight to near darkness
- Color night vision, no active illumination
- Replace or support conventional cameras
- Detect unilluminated pedestrians and cyclists
- Enable reliable automated parking at night
- Maintain safety during sunrise, sunset & glare

Technology:

- Single-photon avalanche diode (SPAD): sense every photon
- <1W power draw for complete camera module
- Up to 5.4 MP for far & near-sensing applications
- 136 dB in single exposure: no signal-to-noise drops
- Other specs: 120 FPS, 8.2 ms exposure, GMSL2 SerDes, RAW14

Low Light Infrared Solution for Satellite Camera



**SOP
2029**

Benefits & Features:

RGB-IR camera solution provides increased night-time performance while maintaining color accuracy for human drivers and machine vision:

- Sensitive to invisible IR illumination: exceptional visibility on the dark sides of the vehicle
- Daytime Color Performance: no sacrifices on color accuracy
- Platform Integration: plug & play into existing RGB platforms & CV stacks like Nvidia and Qualcomm

Technology:

- Infrared-enhanced imaging without compromising daytime performance
- Up to 8 MP for far & near-sensing applications
- Remosaicing algorithm reconstructs authentic RGB images
- Complete image processing chain preserving image quality
- Other specs: 60 FPS, 130 dB, GMSL2, RAW12

Select to open the corresponding one-pager

Democars 1, 2, 4 & 7 are located on the Outside Visitor Parking Area

1

Xelve Park - USS and Vision based Advanced Parking and Human Vision

2

Xelve Park - Radar Vision Parking

4

Deviceless Access Extended

7

Cockpit HPC & SDV Solutions

Democars 3, 5, & 6 are located in the “Salle des Comités” (Showroom)

3

Cabin Sensing

5

Mirrorless 3D Augmented Reality HUD

6

Ac2ated Sound Technology

Xelve Park

USS and Vision Based Advanced Parking and Human Vision



In SOP

Benefits & Features:

- Cost efficient technology solution for advanced (Level 0 - 2) parking functions including Human Vision and Park Augmentation.
- Low level fusion of neural network processed camera (SVC) and Ultrasonic sensors (CUS).

Technology:

- 4 x SVC330 - Surround View Camera
- 1 x ADC544 - ECU
- 12 x CUS320
- Human Vision SW Gen 2 on embedded ECU
- Transparent Vehicle
- Augmented Reality for Parking Functions for exceptional user experience
- Rearview, 2D Topview and 3D Surround View
- Parking SW Gen 1 - On embedded ECU
- Scene: Parallel / Perpendicular / Angled Parking / between vehicles / User Defined Parking

Xelve Park

Radar Vision Parking



**Under
Development**

Benefits & Features:

- **Cost efficient solution** for advanced SAE L2 / L4 parking functions that makes the use of **Ultrasonic Sensors obsolete.**
- **Reduced components** and improving aesthetic vehicle design,
- **Design Freedom** and Simplification
- **Ease of Use:** A clean, intuitive UI helps users quickly understand how to take advantage of the Parking functionality
- Higher performance due to **next generation perception** concepts
- Portable on central architectures, de-couples HW from SW

Technology:

- Our AI-powered early slot detection and Radar Vision Parking deliver superior performance while reducing sensor complexity and enabling design freedom
- New high bandwidth mode (79GHz) gives higher resolution in the near range, elevation measurement capacity
- Curb-stone height estimation, high resolution maps, along with normal mode(77GHz) features.
- With our white-box approach, OEMs can choose from software-only offerings to complete system solutions including sensors, accelerating time-to-market and lowering development costs.

Cabin Sensing Evaluator

Comprehensive vehicle setup for Cabin Sensing evaluation



SOP
2029 to 2035

Benefits & Features:

- Fast algorithm development loop through high-quality, labeled datasets.
- Consistent benchmark environment for suppliers, partners, and internal teams.
- Provides the basis to test and demonstrate new use-cases (stress detection, occupancy classification, health monitoring).
- Provides a scalable reference architecture for current and future solutions.

Technology:

- Instrumented test vehicle integrating multiple in-cabin sensing modalities
 - 60 GHz Radar,
 - UWB in radar mode,
 - RGB/NIR cameras,
 - Capacitive seat sensors.
- Recording environment enabling synchronized capture of raw data streams (vehicle bus, sensor data, video, ground-truth references).

Intelligent Vehicle Experience

Deviceless access, surround sensing, external touchless interaction



SOP
2028 (ITO) to
2035 (DLA)

Benefits & Features:

- **Seamless experience:** Secure access with user recognition in all environment conditions
- **Natural Human Vehicle Interaction** through Biometric, posture, gaze... enable touchless vehicle control (eg: **Intuitive Trunk Opening**, intention recognition...)
- **Proactive 360° Security** & Intrusion prevention : Alert when suspicious activity is detected
- Cooperative innovation platform to **jointly** explore & **co-develop** features

Technology:

- **Optimize cost** by reusing existing vehicle sensors : **UWB** , **Cameras**
- **AI based sensor fusion** algorithm
- Intelligent energy management: UWB low consumption sensors for **on-demand wake-up** system
- Security & privacy: multifactor authentication, anti-spoofing up to level 3, RGPD compliance
- Robustness to environment (Camera heating, cleaning...)

Mirrorless 3D Augmented Reality HUD

Enhance your driving reality



Stellantis/AUMOVIO
Innovation phase in
Progress

**SOP
2029**

Benefits & Features:

- Intuitive perception of Driving information
 - navigation,
 - pedestrian warning,
 - adaptive cruise control,
 - lane departure warning,
 - night vision
- Simplified integration (packaging size reduction)
- Cross careline solution

Technology:

- Eye-tracking IR Camera
- Stereoscopic 3D TFT display with local dimming
- Large Image size: 15° x 8°
- High Brightness > 10.000cd/m²
- High Resolution > 70px/°
- Small Packaging size: ~8 liters
- large virtual depth range: 0.6 - 80 meters image distance
- Maturity : TRL6 end of 2026

Ac2ated Sound Technology

Generate sound with structures of the car for interior or exterior



**SOP
2026**

Benefits & Features:

- weight reduction
- less installation space
- improved exterior sound radiation
- new design freedom since no grilles are required
- enables new installation areas on the vehicle

Technology:

- audio actuators for sound generation
- 4-ohm components compatible with any audio amplifier
- 2 types can cover the whole audible frequency range (20Hz – 20 kHz)
- IP6K9K for AS Wide Range

Flexible HPC Solutions for every vehicle architecture



**SOP
2027+**

Benefits & Features:

- Reduced system complexity through smart consolidation of multiple ECUs into centralized body and cockpit HPC platforms
- Lower total cost of ownership through optimized hardware utilization and SW reuse
- Faster time-to-market enabled by production-ready HPC platforms and extensive integration expertise
- Long-term scalability aligned with evolving SDV roadmaps and future vehicle generations

Technology:

- Cockpit and Body HPC solutions based on Android, Linux, QNX, and AUTOSAR.
- Support for both legacy and zonal/SDV architectures
- Cross-domain integration expertise combining cockpit, body, AI, and computer vision functions
- Enhanced safety and convenience through AI-powered perception and cabin intelligence
- Continuous feature evolution supported by modern software-centric architectures